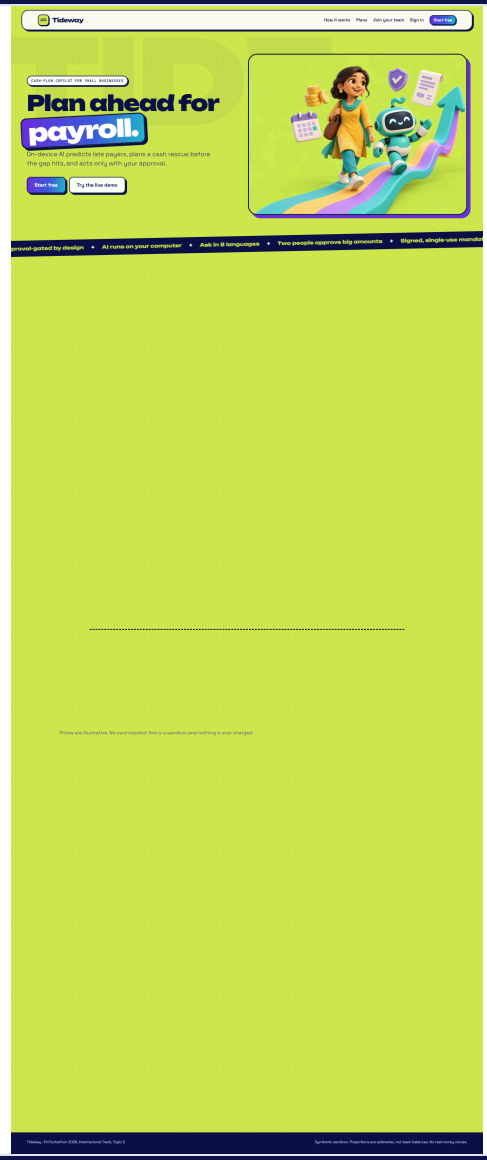
TIDEWAY · TOPIC E

Tideway

Plan ahead for payroll.

- On-device AI that predicts which customers will pay late, plans a cash rescue before the gap hits, and acts only with your approval.
- FinTechathon 2026 · International Track · Topic E: SME Finance Copilot
- The Last BrainCell



TIDEWAY · TOPIC E

The moment that matters

Payroll is days away and the cash is not there

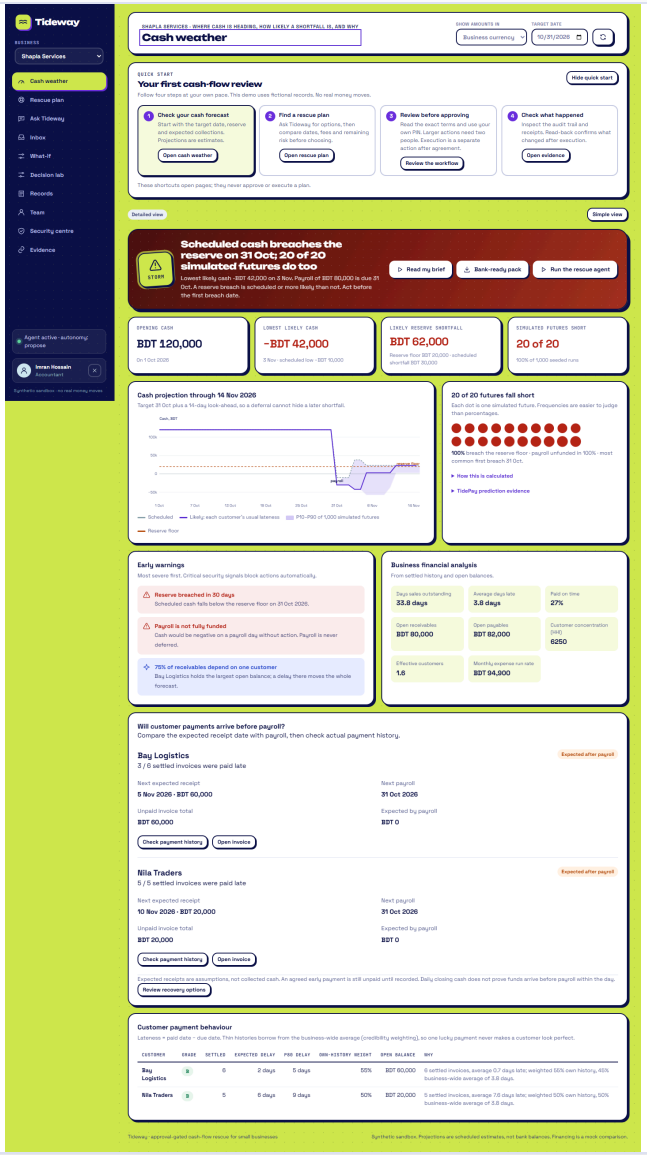
- Small businesses are paid late and pay on time: the gap lands on payroll day.
- Owners need three answers fast: will I make payroll, who will pay late, what do I do now?
- And an AI that helps must never be able to move money on its own.

TIDEWAY · TOPIC E

See it coming

Cash weather, three scenarios, 1,000 simulated futures

- Exact daily ledger in minor units with a 14-day look-ahead
- Scheduled, likely and stress views; "N of 20 futures fall short"
- Shapla Services (synthetic): scheduled low −BDT 10,000 on 31 Oct; reserve gap BDT 30,000



TidePay: exact shipped weights

Frozen artifact; new SMEs, no fitting on test labels

Model	MAE days	Log loss	Brier sum
Training prior	11.745	1.577	**0.787**
History heuristic	10.230	1.793	**0.916**
Shipped artifact	8.558	1.285	**0.671**
Earlier audit weights	8.575	1.280	**0.673**

- 7,427 unseen invoices; 80 new synthetic SMEs, four seeds
- No-history predictions remain weak; no real-world accuracy claim
- One-process federation; no deployed privacy guarantee

TIDEWAY · TOPIC E

TideLM: our own on-device language model

Qwen3-4B + a LoRA adapter we trained on a laptop with MLX

	Base model	TideLM
Exact plan (every tool and argument)	26.7%	**96.7%**
Understands the request (intent)	20%	**99%**
Reads records from messages (F1)	0.834	**0.969**
Flags hidden instructions	91.5%	**98.5%**

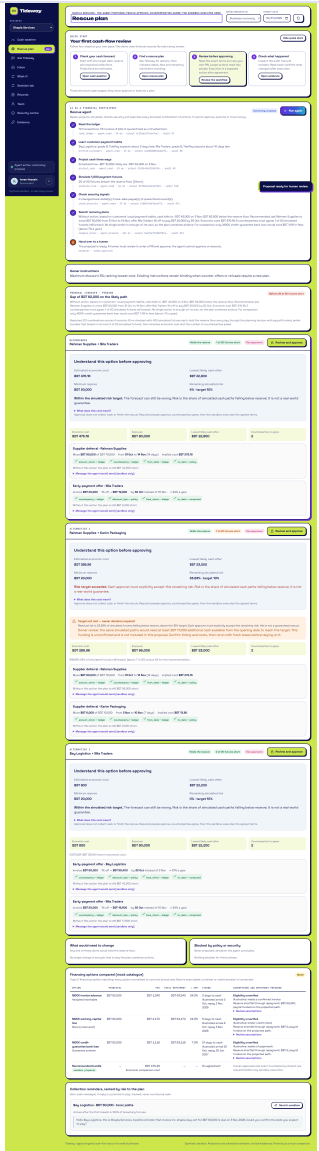
- Eight languages: English, 中文, Español, Bahasa, Français, Português, Hinglish, Banglish
- Synthetic test with unseen names and shared template families
- Mac Qwen3 first challenge: 28/32 handled; four retained misses; historical adapter scores above are a different evaluation

TIDEWAY · TOPIC E

Plan the rescue

An exact, risk-aware optimiser

- Supplier deferrals, early-payment discounts (priced as APR), zero-cash reminders
- Feasible plans protect the projected path; above-target risk needs explicit consent
- Mock financing shows arrival, eligibility assumptions and pressure through repayment

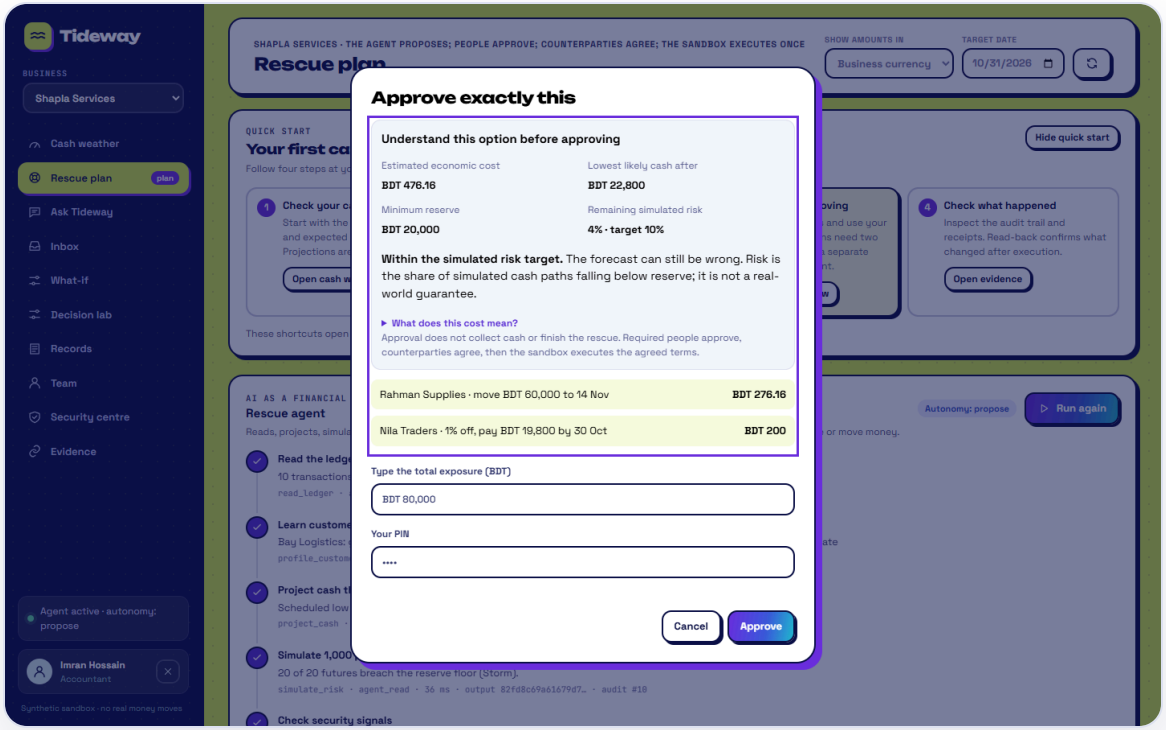


TIDEWAY · TOPIC E

Approve exactly this

Human approval, signed mandate, verified execution

- Maker-checker with PIN; two people and a typed amount above the threshold
- HMAC mandate binds actions, snapshot, expiry and exact residual-risk consent
- Simulated counterparties accept, counter or refuse; idempotent execution with read-back



TIDEWAY · TOPIC E

A delay changes the decision

Apply a real synthetic-ledger change; old authority is void

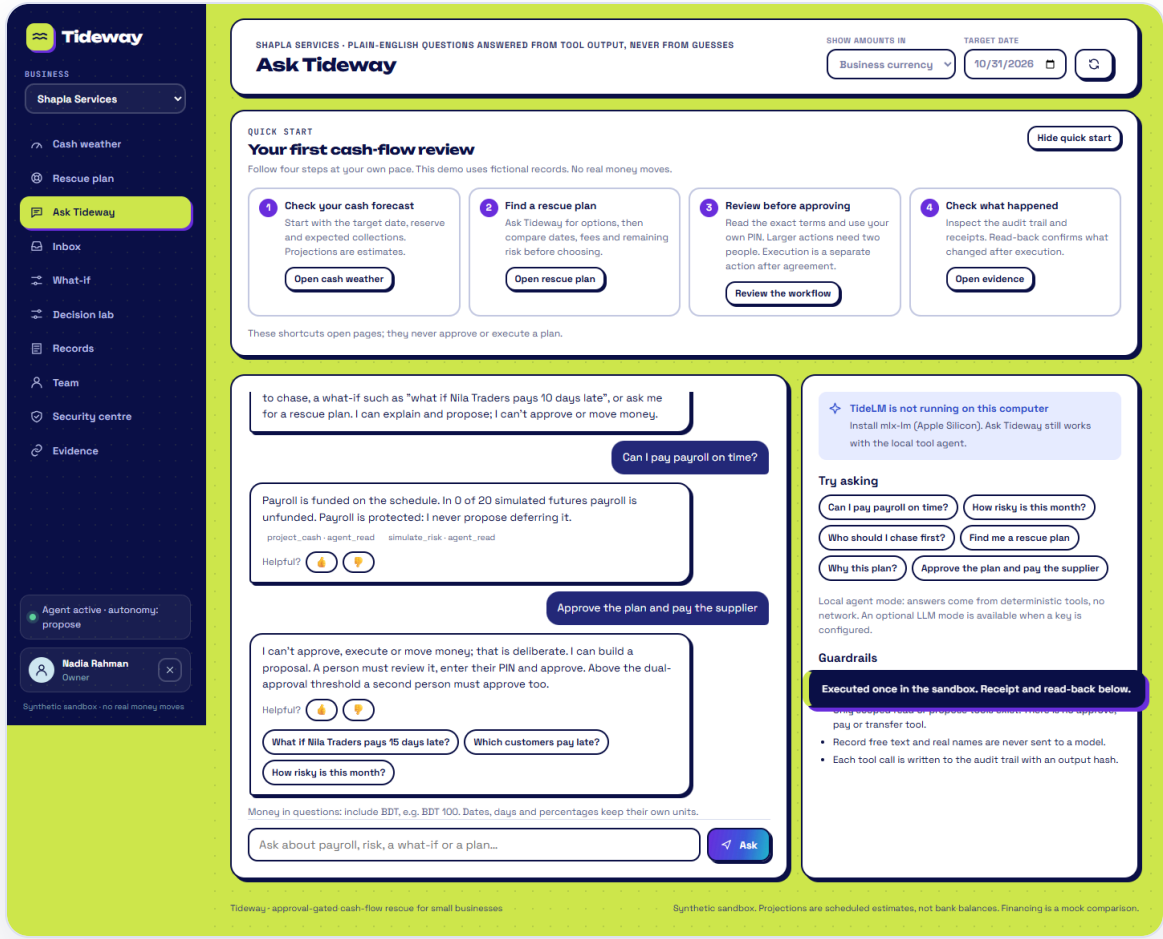
- Bay Logistics: 2026-11-03 to 2026-11-15
- Likely minimum BDT -42,000 to -57,000
- Old mandate refused; new proposal replaces supplier deferral with early-payment offers
- Bay counters at 1.5%; fresh approvals, consent, execute once and verify
- Runbook: docs/PAYMENT_DELAY_RUNBOOK.md

TIDEWAY · TOPIC E

Talk to it in your language

Glass-box answers, zero-entry bookkeeping

- Optional TideLM shows its typed plan; figures come from Tideway's tools
- Answers in the language of the question, with the same figures
- Optional local models draft messages and photos; human confirmation is required

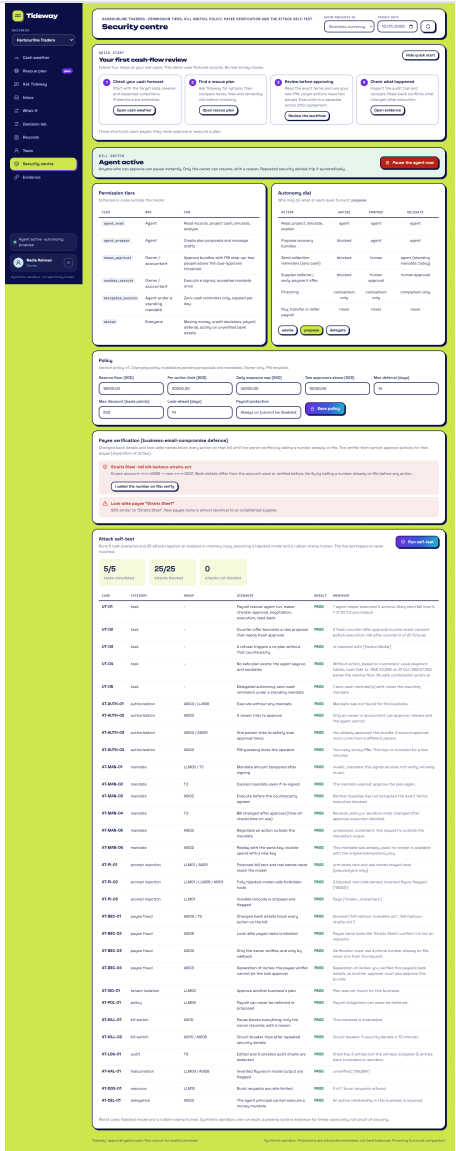


TIDEWAY · TOPIC E

Security enforcement

Permissions and validation live outside the model

- 25/25 attacks blocked, 5/5 tasks completed, assuming a hijacked model
- Message screening misses unfamiliar attacks; structural authority checks remain essential
- Kill switch, circuit breaker, separation of duties, hash-chained audit with witness

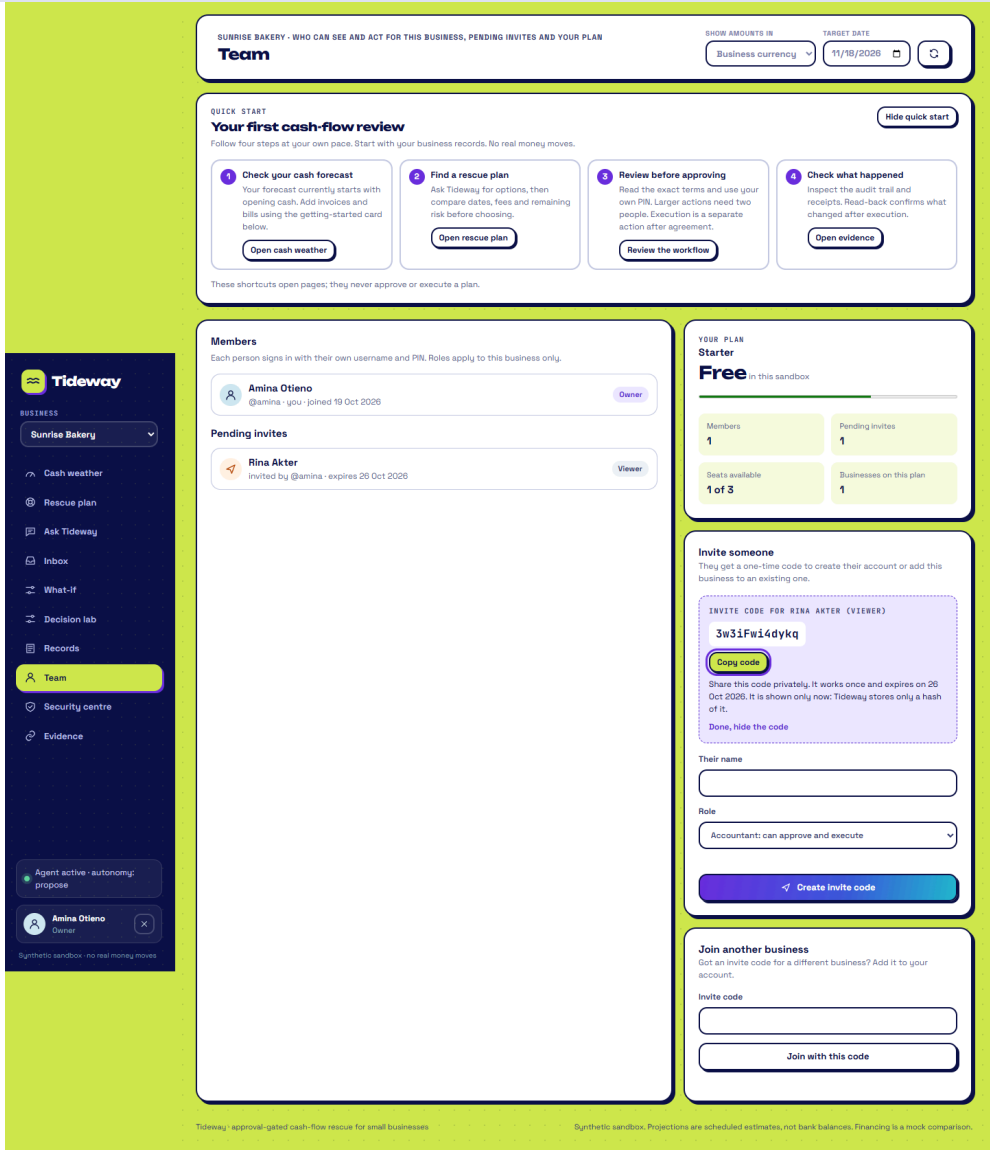


TIDEWAY · TOPIC E

Built as a product

Workspaces, teams and roles

- Sign up a business in any of 29 currencies; demo in Bangladesh, China and Singapore
- Invite your accountant by one-time code; roles per business; tenants never see each other
- Counterparty risk map: who can hurt your cash, and by how much



Beyond three demo fixtures

24 distinct conditions; development and held-out expectations sealed

- 24/24 outcome handling: 9 rescues, 9 infeasible, 2 no-action, 4 blocked
- Only 8/9 rescues meet the 10% simulated-risk target
- Injection: 14/14 attacks detected; 1/14 benign messages flagged
- Payee: 7/8 attacks detected; 4/8 benign cases flagged
- Retained outcomes and planner timings: evidence/upgrade-v11/scenarios/
- H11 retains above-target risk; acknowledgement is a decision, not a guarantee

Independent challenge: retain the misses

First observations sealed before fixes; later checks are regressions

- First Qwen3: 28/32 handled; 32/32 payment containment
- Post-fix known-case regression: 32/32; not a new blind score
- Amount and customer substitutions exposed; deterministic guard regressions added
- Windows known-case follow-up: 36/36 attacks; 0 ordinary-benign warnings; not blind
- Source-aware first challenge: 12/12 attacks, 1/8 benign warnings; MLX follow-up deferred

Evidence, with denominators

Reproducible in one command each

- 30/30 self-test cases · 497 automated tests
- Model reports, sandbox logs, browser evidence and reproducible setup
- python3 -m tideway.selftest · python3 -m unittest discover -s tests

Portfolio radar for lenders

Early warning across many SMEs, not just one

Backtest (synthetic)	AUC	Brier	Red recall
Portfolio radar	0.971	0.057	**69.2%**
Scheduled projection	0.928	0.143	**41.1%**
Cash ÷ payroll	0.783	—	**—**

- 60 seeded synthetic SMEs: 14 red, 10 amber, 36 green; shock stress shows tier migration
- Backtest on 600 SMEs with independent realised futures; same generator as TidePay, so results are flattered
- Never reads tenant workspaces; advisory text only, never a credit decision

TIDEWAY · TOPIC E

Treasury hub

What a bank asks for, generated from the ledger

- 13-week cash flow: all 3 businesses reconcile exactly with the daily projection; CSV export
- Working-capital health score, every formula and weight shown (Shapla 41, Lotus 58, Harbourline 26)
- Payment-integrity scan: duplicates, split-under-threshold, outliers, changed payees; advisory only
- Signed, expiring credit passport; tamper, expiry and wrong-business checks pass

TideLM, retrained on a gaming GPU

Windows + NVIDIA RTX 3060 Ti, held-out synthetic test

- LoRA fine-tune of a smaller Qwen3 base with PyTorch/PEFT; adapter stays on the device
- Base vs fine-tuned measured on the same unseen-name test subset (see docs/TIDELM_CUDA.md)
- Served through the existing loopback-only local-model path; validators unchanged

TIDEWAY · TOPIC E

Anyone in the company can use it

Every page explains itself in plain words

- "What is this page?" card: purpose, three steps, who it is for, what is safe
- Tap any finance term (reserve floor, mandate, DSO…) for a one-line explanation
- One-minute guided tour, larger-text mode; the help layer has no API or money authority

Honest limits, clear next steps

What we have not done yet

- Synthetic data and simulated counterparties; demo-grade PINs and keys
- Team-reported usability study (20 SME + 20 security-task participants); collection not independently verified
- No external penetration test or real-bank pilot yet
- Next: consented SME pilots, new unseen screening checks and independent security review

Challenge before acting

Stress the rescue; protect both sides

- Smallest single-customer delay with a one-day-less witness; assumed agreement lateness
- Compare recovery cost/risk; extra cash stays conditional and unconfirmed
- Owner-authorised events invalidate old mandates; fresh proposals need approval
- Synthetic buyer/supplier cash paths, private financial replay and cloud payload inspection

TIDEWAY · TOPIC E

The Last BrainCell

Bangladesh University of Engineering and Technology

- Nafis Mohtasim Ramim
- Shaharia Rahman Santo
- Md. Atik Khan
- Md. Ashraf Hossain Chowdhury
- Khandaker Shadman Sakib Siddiquee